

A Flexible Vibrotactile Feedback System for Rapid Prototyping

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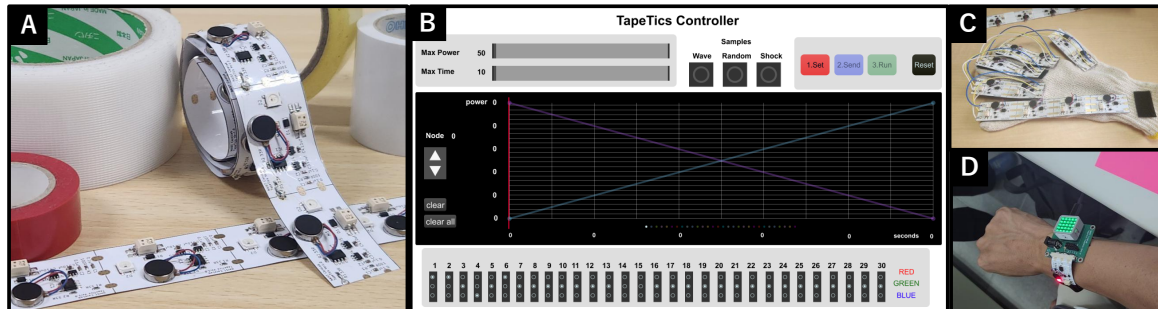


Figure 1: (A) Appearance of Tape-Tics; (B) GUI controller; (C) Example of a haptic glove made by attaching Tape-Tics to a work glove; (D) Vibration alarm wristband created by a participant

ABSTRACT

Tape-Tics is a flexible, modular vibrotactile feedback system integrating LEDs and vibration motors. Users can adjust vibration intensity, LED colors, and configurations via Bluetooth Low Energy (BLE), enabling operation based on predefined settings. Built with flexible printed circuit (FPC) technology, Tape-Tics adapts to curved surfaces, enhancing its versatility in education, research, and art. In a hands-on workshop, participants assembled and operated Tape-Tics to explore vibrotactile technology. Using the GUI controller, they configured vibration patterns and developed unique applications. Surveys showed participants' understanding of haptics improved significantly, and they rated Tape-Tics highly as an educational tool.

Index Terms: Haptic interfaces, Flexible printed circuits, Rapid prototyping, Wearable devices, Educational technology.

1 INTRODUCTION

As VR becomes increasingly popular, tactile feedback has gained attention for enhancing immersive experiences. Technologies such as linear resonant actuators (LRA), and eccentric rotating mass (ERM) motors are commonly used in devices like gaming consoles and mobile phones. Educational institutions are incorporating haptics into STEM programs, but there is a lack of instructional kits focused solely on vibrotactile feedback.

While toolkits, such as the TECHTILE Toolkit [5] offer frameworks for haptic feedback, they often require significant technical expertise for expansion. Flexible devices such as SensorTape [2] are designed with pre-mounted components on flexible substrates, making them easier to use without complex assembly. Reconfig-

urable modular haptic devices including SoftVibe [4] and VibroForge [3] allow users to combine and customize the shape, size, and type of tactile stimulation easily. Modular devices like ChainFORM [6] provide flexibility but are less accessible due to intricate assembly requirements. However, they are often limited to specific body parts, and complex wiring makes prototyping with multiple vibrators challenging.

To address these challenges, we propose Tape-Tics, a modular, tape-like device inspired by LED strip lights (Figure 1 (a)). Tape-Tics features small vibrators integrated with NeoPixel-controlled LEDs, mounted on a flexible printed circuit (FPC) board. Users can cut Tape-Tics to any desired length, attach it to various surfaces, and receive both visual and tactile feedback. Each node's vibrator and LED can be independently controlled, with vibration intensity adjustable in 256 levels. Leveraging the NeoPixel library ensures compatibility with extensive online resources, enhancing scalability.

To simplify usage, we provided a GUI controller that runs on a PC, enabling users to adjust settings and operate the device via BLE without advanced programming skills. This user-friendly approach lowers the barrier for non-experts.

In a workshop for high school students, participants assembled and customized Tape-Tics to create unique projects, such as vibration alarm wristbands and hats with tactile notifications. Surveys revealed that Tape-Tics effectively enhanced participants' understanding of haptics and was highly rated as an educational tool.

2 DESIGN AND IMPLEMENTATIONS

The hardware consists of two main components: Tape Nodes and a Master Node. Tape Nodes are built on FPC board, ensuring adaptability to curved surfaces. Each node includes a NeoPixel-compatible IC (WS2811), a vibration motor, and an RGB LED, all independently controllable. Power is supplied via a 5V source with voltage regulators for stable operation. The design allows users to cut and connect nodes easily, simplifying customization. The Master Node includes an M5Stack Atom Matrix for control, enabling wireless communication via BLE. It integrates GPIO pins for compatibility with external sensors and connects to Tape Nodes through an connector. The modular design supports various applications while remaining compact and user-friendly.

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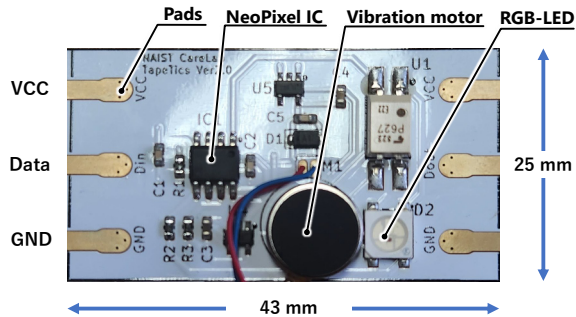


Figure 2: Components of the Tape Node

Tape-Tics uses serial communication for seamless control of vibration and LED functions. Each WS2811 IC processes its own data and relays the remaining data downstream, eliminating complex wiring. The system leverages the FastLED library for efficient LED control, offering precise timing and compatibility with platforms such as Arduino and ESP32. A GUI controller developed with Max8 enables users to configure Tape-Tics without coding, providing an intuitive interface that allows non-programmers to explore haptic technology easily (Figure 1 (b)).

3 TECHNICAL EVALUATION

The manufacturing cost of a Tape Node is approximately \$2.697 per unit. For the Master Node, the total cost per PCB was \$21.95. The power consumption per Tape Node is 128 mA, including the vibrator (80 mA) and LED (48 mA). The M5Stack controller consumes 80 mA (0.4 W). Using a copper plane with a maximum current rating of 10 A, up to 78 nodes can be operated simultaneously. With a mobile battery rated at 3.6 V, 3350 mAh, and an output current of 2.4 A, the maximum number of nodes is 18, which provides an operational runtime of approximately 1.39 hours.

4 USER STUDY: TAPE-TICS WORKSHOP

4.1 User Evaluation

A hands-on workshop was conducted with high school students to evaluate the effectiveness and versatility of the device and its GUI controller. Ethical approval for this study was obtained from the Institutional Review Board of our organization. The workshop involved 10 male students aged 15–18 ($M = 16.3$, $SD = 0.9$), including two Mechanics majors and eight Electronics majors. The evaluation was based on quantitative data collected through 5-point Likert scale surveys, as well as qualitative insights gathered from interviews, sticky notes, presentation materials, recorded videos, audio recordings, and open-ended survey responses. Furthermore, key themes were identified during the analysis process [1].

4.2 Analysis and Results

In the pre-workshop surveys, the mean score for Q1, “How much do you know about haptic technology?” was 2.3, indicating that participants’ prior knowledge was generally low. In contrast, post-workshop responses showed an increase in understanding, as reflected in Q3, “Did you understand the tactile lecture?” which received a mean score of 4.1.

Additionally, participants expressed high satisfaction with the overall workshop experience, as shown in Q6, “How much did you enjoy the workshop?” which received a mean score of 4.9, indicating that most participants found the workshop enjoyable. Most participants found the workshop engaging, particularly the hands-on activities and teamwork. They enjoyed exploring Tape-Tics’ flexibility and creating unique designs.

Regarding the usability of the Tape-Tics system, Q7, “Was the controller operation easy to understand?” received a mean score of 4.4, indicating that the controller was generally easy to use. The flexibility and ease of installation of the device, as assessed in Q8, “How easy was the device to flex, cut, and install?” received a positive rating, with a mean score of 4.2. However, inserting and removing tape from the connector received a lower score of 3.7 in Q9, “Was it easy to insert and remove the tape from the connector?” suggesting that this area requires improvement.

Project completion and implementation smoothness were rated moderately, with mean scores of 3.7 for Q10, “How much of the work is complete?” and 3.8 for Q11, “Did the implementation go smoothly?”, reflecting some technical challenges during the project. Participants suggested several improvements, including more precise frequency and color customization, waveform import/export features, and adjustments to the size and weight of the device for better usability.

5 CONCLUSION AND FUTUREWORK

In conclusion, Tape-Tics offers a versatile, flexible, and user-friendly system for providing vibrotactile feedback. The hands-on workshop demonstrated its effectiveness in teaching haptic concepts and fostering creative applications, such as VR haptic gloves and wristwatch-type devices (Figure 1 (c) (d)). The positive feedback and success of the workshop confirm Tape-Tics’ potential as both an educational tool and a research device.

Future versions will include plug-and-play sensors compatible with the Grove system, which will simplify sensor integration. Additionally, feedback indicated that vibration strength was insufficient for some applications. Future iterations may incorporate more powerful motors, such as LRA, to address this.

While the current GUI is effective for beginners, greater platform compatibility, particularly with smartphones, will be prioritized to improve accessibility. The issue of incorrect connector insertion will be addressed by adding visual guides. Challenges with attaching Tape-Tics modules to objects or body parts will be resolved by refining adhesive properties or developing specialized attachments.

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